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1.grep

Definition

The grep command is used to search for text or patterns inside files. It prints the lines that match the search pattern.

Usage / Formula `grep [options] pattern file`

Examples

```
Search for a word in a file
grep "root" /etc/passwd
Ignore uppercase and lowercase differences
grep -i "linux" notes.txt
Show line numbers
grep -n "error" log.txt
```

2.awk

Definition

The awk command is used to process and analyze text files. It is commonly used to display specific columns or manipulate text.

Usage / Formula `awk 'pattern {action}' file`

Examples

```
Print the first column
awk '{print $1}' names.txt
Print the first and third columns
awk '{print $1, $3}' data.txt
Print lines that contain a word
awk '/admin/ {print}' users.txt
```

3.sed

Definition

The sed command is used to search, replace, and edit text in a file or command output.

Usage / Formula `sed 'command' file`

Examples

```
Replace a word
sed 's/cat/dog/' pets.txt
Replace all occurrences in a line
sed 's/apple/orange/g' fruits.txt
Display only specific lines
sed -n '1,5p' notes.txt
```

4. Pipe (|)

Definition

The pipe operator (|) sends the output of one command directly into another command.

Usage / Formula `command1 | command2` **Examples**

```
Send cat output to grep
cat names.txt | grep "John"
Send ls output to sort
ls | sort
Count lines using wc
cat file.txt | wc -l
```

5. Redirect Output (>)

Definition

The > operator saves the output of a command into a file. If the file already exists, it will overwrite the content.

Usage / Formula `command > file`

Examples

```
Save directory list
ls > files.txt
Save users from /etc/passwd
cat /etc/passwd > users.txt
Save sorted data
sort names.txt > sorted_names.txt
```

6. Append Output (>>)

Definition

The >> operator appends the output of a command to the end of a file without deleting the existing content.

Usage / Formula `command >> file`

Examples

```
Append current date
date >> log.txt
Append directory contents
ls >> files.txt
Append system users
cat /etc/passwd >> users.txt
```